

Lecture 1

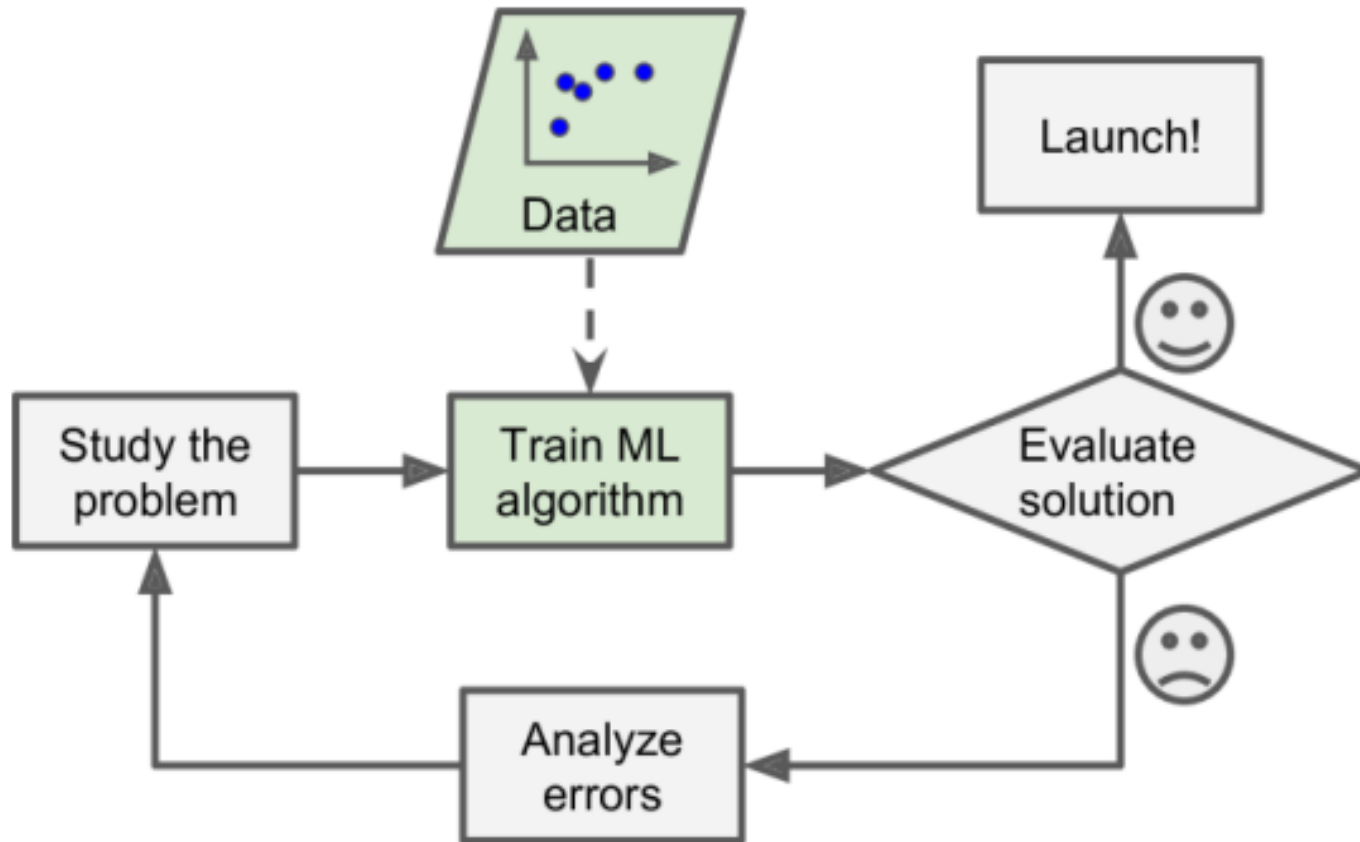
Introduction to Machine Learning

- ☐ What is machine learning?
- ☐ History and relationships to other fields
- ☐ Applications of machine learning
- ☐ Types of machine learning techniques
- ☐ Overview of Data mining and KDD process:
- ☐ Prediction vs. Description modeling

Definition of Machine Learning (ML)

- ❑ ML is a field of computer science with the help of which computer systems can provide sense to data in much the same way as human beings do.
- ❑ It extracts patterns out of raw data by using an algorithm or method.
- ❑ It learns from experience without being explicitly programmed or human intervention.

Machine Learning Approach

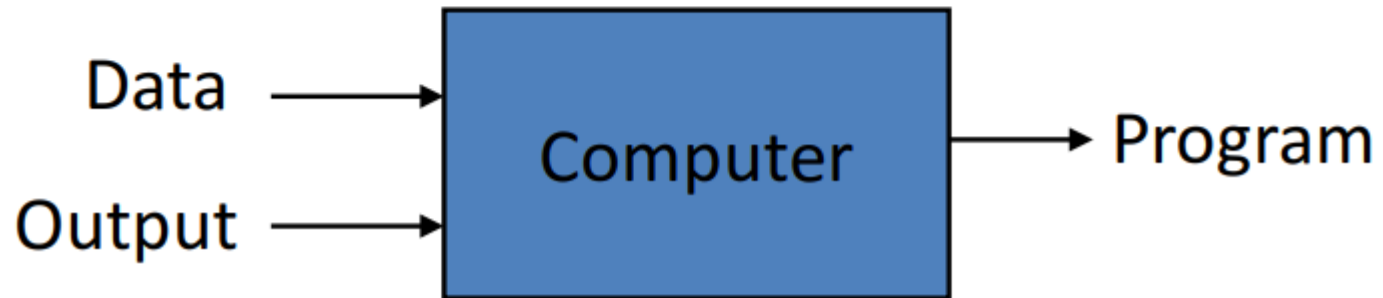


Traditional Programming vs. Machine Learning

Traditional Programming



Machine Learning



History of Machine Learning

- ❑ Machine learning grew out of the quest for artificial intelligence.
- ❑ Already in the early days of AI as an academic discipline, some re-searchers were interested in having machines learn from data then termed "neural networks“.
- ❑ However, an increasing emphasis on the logical, knowledge-based approach caused a rift between AI and machine learning.

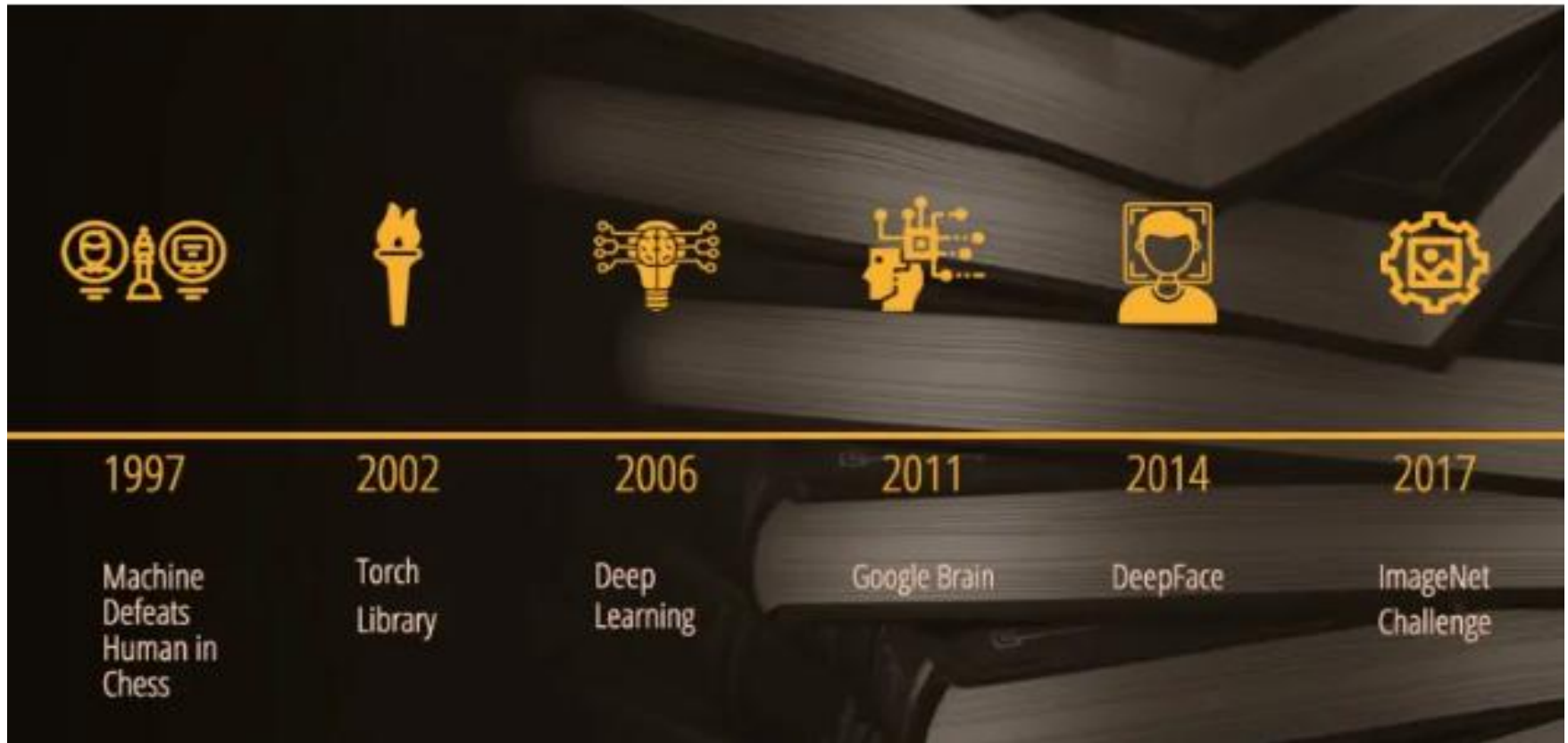
History of Machine Learning

- ❑ Machine learning, reorganized as a separate field, started to flourish in the 1990s.
- ❑ The field changed its goal from achieving artificial intelligence to tackling solvable problems of a practical nature.
- ❑ It shifted focus away from the symbolic approaches toward methods and models borrowed from statistics and probability theory.

Early History of Machine Learning



History of Machine Learning



Present: State-of-the-art Machine Learning

- ❑ Machine learning is used in many different fields, from fashion to agriculture.
- ❑ Machine Learning algorithms are able to learn patterns and relationships between data, find predictive insights for complex problems and extract information that is otherwise too difficult to find.
- ❑ Today's Machine Learning algorithms are able to handle large amounts of data with accuracy in a relatively short amount of time.

Machine Learning and Data Mining

- ❑ Both employ the same methods and overlap significantly.
- ❑ They can be roughly distinguished as follows:
 - ✓ Machine learning focuses on prediction, based on known properties learned from data.
 - ✓ Data mining focuses on the discovery of (previously) unknown properties in the data.
- ❑ Machine learning also has intimate ties to optimization:

Types of Machine Learning

- ❑ Classification of ML in broad categories based on:
 - ✓ Whether or not they are trained with human supervision
(supervised, unsupervised, semi-supervised, and Reinforcement Learning)
 - ✓ Whether or not they can learn incrementally on the fly
(online versus batch learning)

Types of Machine Learning

1	Supervised Learning	✓ Deals with labeled datasets ✓ Two processes ✓ Classification ✓ Regression
2	Un-supervised Learning	✓ It uses unlabeled data ✓ No idea which types of results are expected. ✓ Two types ✓ Clustering ✓ Dimensionality reduction
3	Semi-supervised Learning	✓ Known as hybrid learning and it lies between supervised and unsupervised learning. ✓ Combination of labeled and unlabeled data.
4	Reinforcement Learning	✓ There are no training data sets. ✓ An example of reinforcement learning problem is playing game.

Supervised Machine Learning

- ❑ In supervised learning the machine experiences the examples along with the labels or targets for each example. The labels in the data help the algorithm to correlate the features.
- ❑ It used labeled dataset (Dataset with Class label)
- ❑ Example **classification** and **Prediction (regression)**
 - ✓ Regression: output value is continuous.
 - ✓ Classification: output value is discrete or “categorical”

Unsupervised Machine Learning

- The dataset used is not “labeled”
- Extract the pattern (cluster) from the available data.
- Example Clustering and Association Rule Discovery
 - ✓ **Clustering**: finding “groups” in the data, according to a similarity measure
 - ✓ **Association Rule Mining**: Discover rules that describe large portions of your data, such as people who buy X also tend to buy Y.

Semi-Supervised Learning

- ❑ Hybrid learning architectures combine supervised and unsupervised learning components.
- ❑ In semi-supervised learning, training data is split into two. A small amount of labeled data and a larger set of unlabeled data. In this case, the model uses labeled data as an input to make inferences about the unlabeled data, providing more accurate results than regular supervised-learning models.
- ❑ This approach is gaining popularity, especially for tasks involving large datasets such as image

Reinforcement Learning

- ❑ Reinforcement learning (RL) is concerned with how a software agent (or computer program) ought to act in a situation to maximize the reward.
- ❑ RL attempts to determine the best possible path it should take in a given situation.
- ❑ This machine learning method is mostly used in robotics and gaming. Video games demonstrate a
- ❑ clear relationship between actions and results, and can measure success by keeping score.

The Machine Learning Process



The Machine Learning Process

❑ Problem Framing

- **Decide what kind of problem** are you trying to solve.
 - ✓ Develop a model to classify emails as spam or not spam
 - ✓ Develop a model to classify tumor as malignant or benign,
 - ✓ Develop a model to classify English language to Amharic.
 - ✓ Pattern discovery that shows the association of items in supermarket shelves.

The Machine Learning Process

- ❑ Data Analysis
- ❑ Collect the dataset available for developing the model.
 - ✓ Data visualization of features,
 - ✓ Handling missing data, handling categorical data,
 - ✓ Encoding class labels,
 - ✓ Normalization, and standardization of features,
 - ✓ Feature engineering,
 - ✓ Dimensional reduction,
 - ✓ Data partitioning into training, validation and testing sets, etc.

The Machine Learning Process

- ❑ Model Building
 - ❑ This is the core part of machine learning by selecting appropriate ML algorithms.
 - ✓ linear regression, logistic regression, KNN, SVM, K-means, apriori algorithm, time series analysis, etc.
 - ✓ The model developed from the training dataset
 - ✓ Performance is evaluated using the testing dataset
- ❑ Performance Evaluation
 - ✓ Accuracy, Precision, Recall
 - ✓ Support, Confidence

The Machine Learning Process

- ❑ Application
- ❑ The final machine learning model is put into production to assist decision-making in
 - ✓ Increasing productivity
 - ✓ Deciding if a bank should approve credit to a borrower, etc.

Challenges in Machines Learning

- ❑ The major challenges that ML is facing currently are:
 - ✓ No clear objective for formulating business problems
 - ✓ Quality of dataset.
 - ✓ Time-Consuming task:
 - ✓ Lack of specialist persons:.
 - ✓ Issue of overfitting & underfitting
 - ✓ Curse of dimensionality
 - ✓ Difficulty in deployment

Applications of Machines Learning

- ☐ Sentiment analysis
- ☐ Image Recognition
- ☐ Virtual Assistants
- ☐ Product Recommendations
- ☐ Medical Diagnosis
- ☐ Network intrusion detection
- ☐ Weather forecasting and prediction
- ☐ Stock market analysis and forecasting
- ☐ Speech recognition
- ☐ Customer segmentation
- ☐ Object recognition
- ☐ Fraud detection

Assignment-1

- Take a research paper (thesis) of a machine learning application and write the following
 - ✓ Title of the research
 - ✓ Objective of the research
 - ✓ From where the dataset collected
 - ✓ What preprocessing techniques applied on the dataset
 - ✓ What a machine learning technique(s) applied on it
 - ✓ What a performance result has been obtaine

Revision of Basic Statistics

- **Reading Assignment**

- Mean, Median, Mode
- Max, Min, Range
- Variance, Standard Deviation
- Quartiles, Interquartile range
- Histogram, Scatter plot
- Symmetric data, Positively skewed, Negatively skewed

Python Library (Pandas)

- ❑ It is one of the most famous Python libraries and used for importing and managing the datasets.
- ❑ It is an open-source data manipulation and analysis library.
- ❑ It will be imported as below:

import pandas as pd

Python Library (Numpy)

- ❑ Numpy Python library is used for including any type of mathematical operation in the code.
- ❑ It is the fundamental package for scientific calculation in Python.
- ❑ It also supports to add large, multidimensional arrays and matrices. So, in Python,
- ❑ Imported as:

import numpy as np

Python Library (Matplotlib)

- ❑ It is a Python 2D plotting library.
- ❑ We need to import a sub -library pyplot.
- ❑ This library is used to plot any type of charts in Python for the code. It will be
- ❑ imported as below:
- ❑ `import matplotlib.pyplot as plt`

Python Library (Pandas)

❑ Python toolboxes/libraries for data processing:

- ✓ NumPy
- ✓ SciPy
- ✓ Pandas

❑ Visualization libraries

- ✓ matplotlib
- ✓ Seaborn

❑ Machine learning & deep learning

- ✓ Scikit-learn ◦
- ✓ Tensorflow/Pytorch/Theano